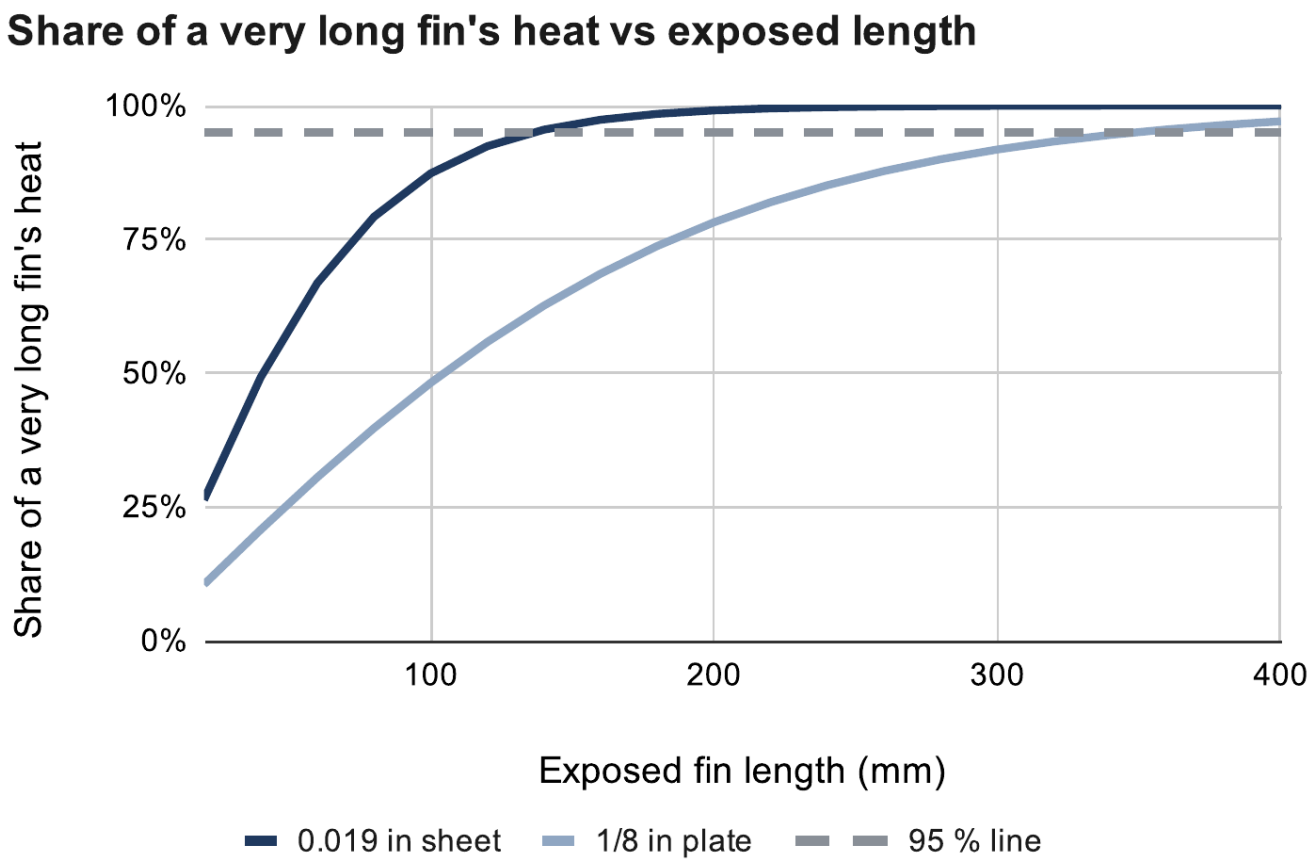
Appendix

Safe State and Failure Modes

Event	What happens	Mitigation
PAUSE, or a fault the software detects	Every station holds position with torque on	Nothing moves again without approval
Wi-Fi link lost	Stations stop and hold on their own clocks	Resume needs approval
Power loss or emergency stop	Torque is lost and the claw can fall (1.5 kg from about 2 m)	Nobody underneath; claw parked low when idle; brake planned
Controller reset or brown-out	Driver disabled at boot, so torque is lost	Same as power loss
Driver over-temperature	The A4988 shuts its outputs off	Hold test watches the driver; fallback 1.35 A

Heat Fin Length



- 0.019 in prototype reaches 95 % of its capacity by 136 mm
- 1/8 in design keeps gaining to 349 mm
- Built at 200 mm, limited by ceiling clearance

Calculated (calc sheet: Heat fin)

Fin length and efficiency

$m = \sqrt{(2h / (k\,t))}$, $L_{95} = 1.83 / m$

$\eta = \tanh(mL) / (mL) =$ **0.74** (1/8 in)

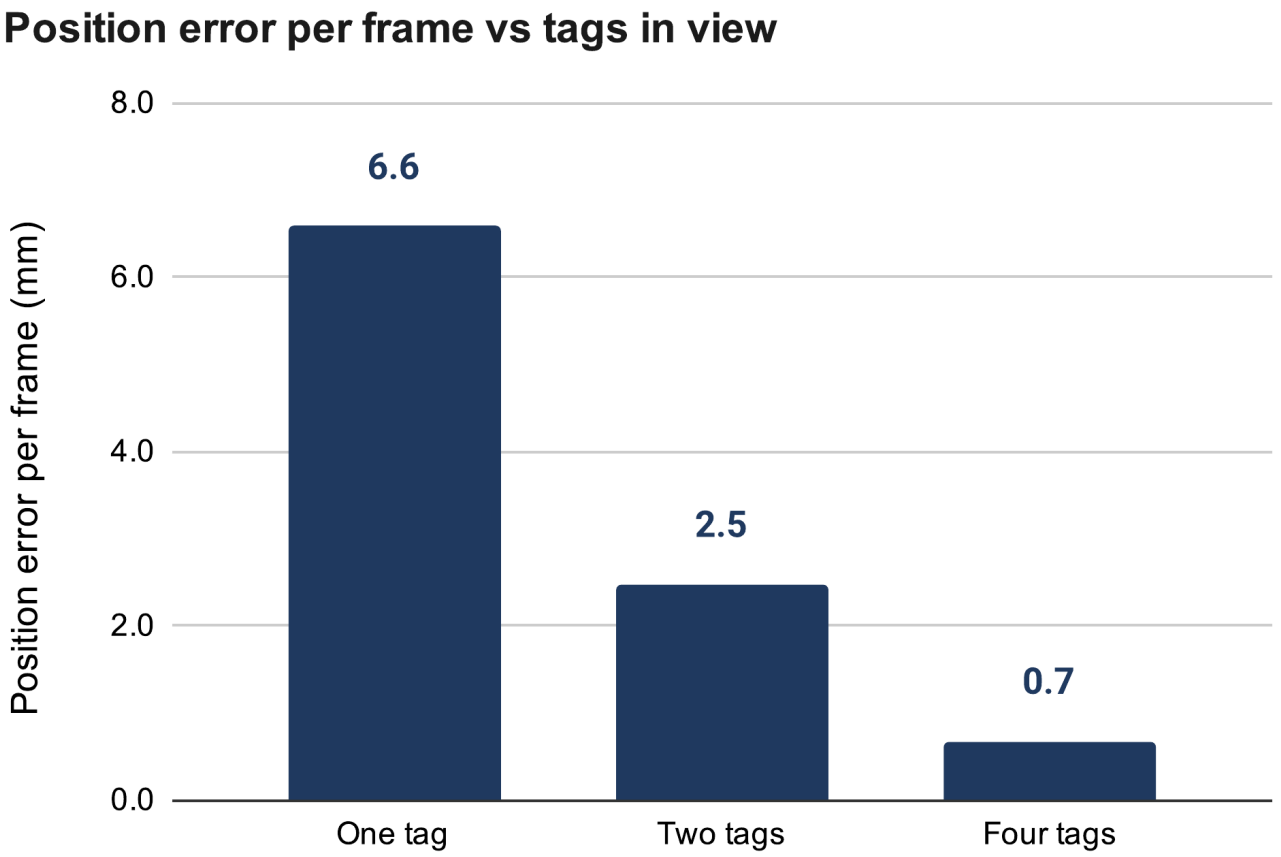
Fin Sizing for 4 W

Painted 6061, 44 mm wide	Mass	Shortest for ≤ 60 °C
0.019 in (prototype)	14 g	None (2.0 W max)
1/16 in	47 g	None (3.6 W max)
0.090 in	68 g	210 mm or none
1/8 in (design)	94 g	145–185 mm
3/16 in	141 g	125–140 mm
1/4 in	189 g	115–125 mm

All 4.05 W through the fin, 23 °C room, paste contact. Calculated.

- If the fin alone had to carry all 4.05 W
- 0.019 in sheet: not possible at any length
- 1/8 in plate: 145–185 mm

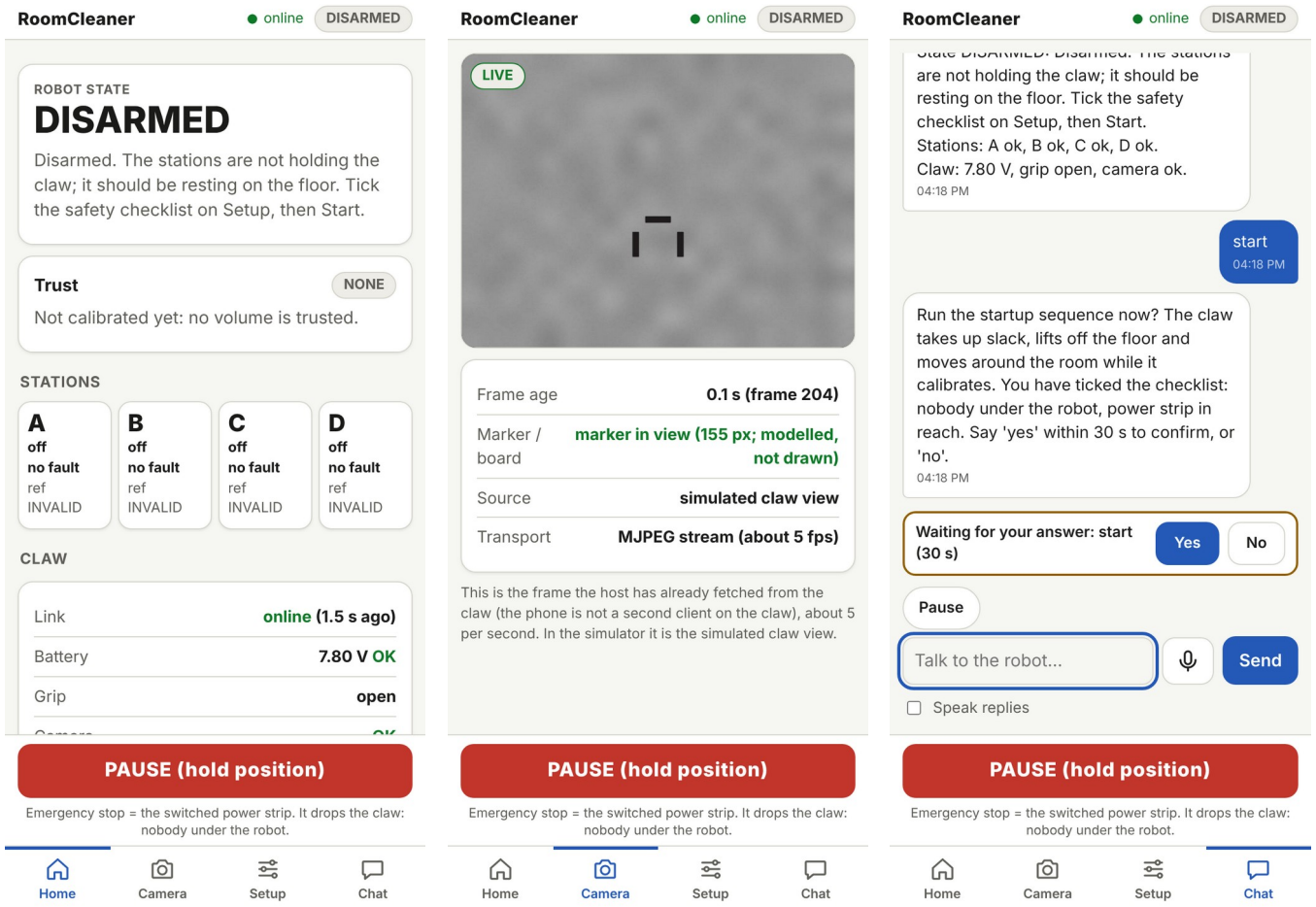
Floor Board Accuracy



- Camera pose error per frame, best case: 6.6 mm with one tag, 0.7 mm with four
- This is the camera's view, not the cable length
- After the fit, cable lengths are known to about 12–16 mm (simulated)

Calculated (calc sheet: Floor board)

Operating Sequence



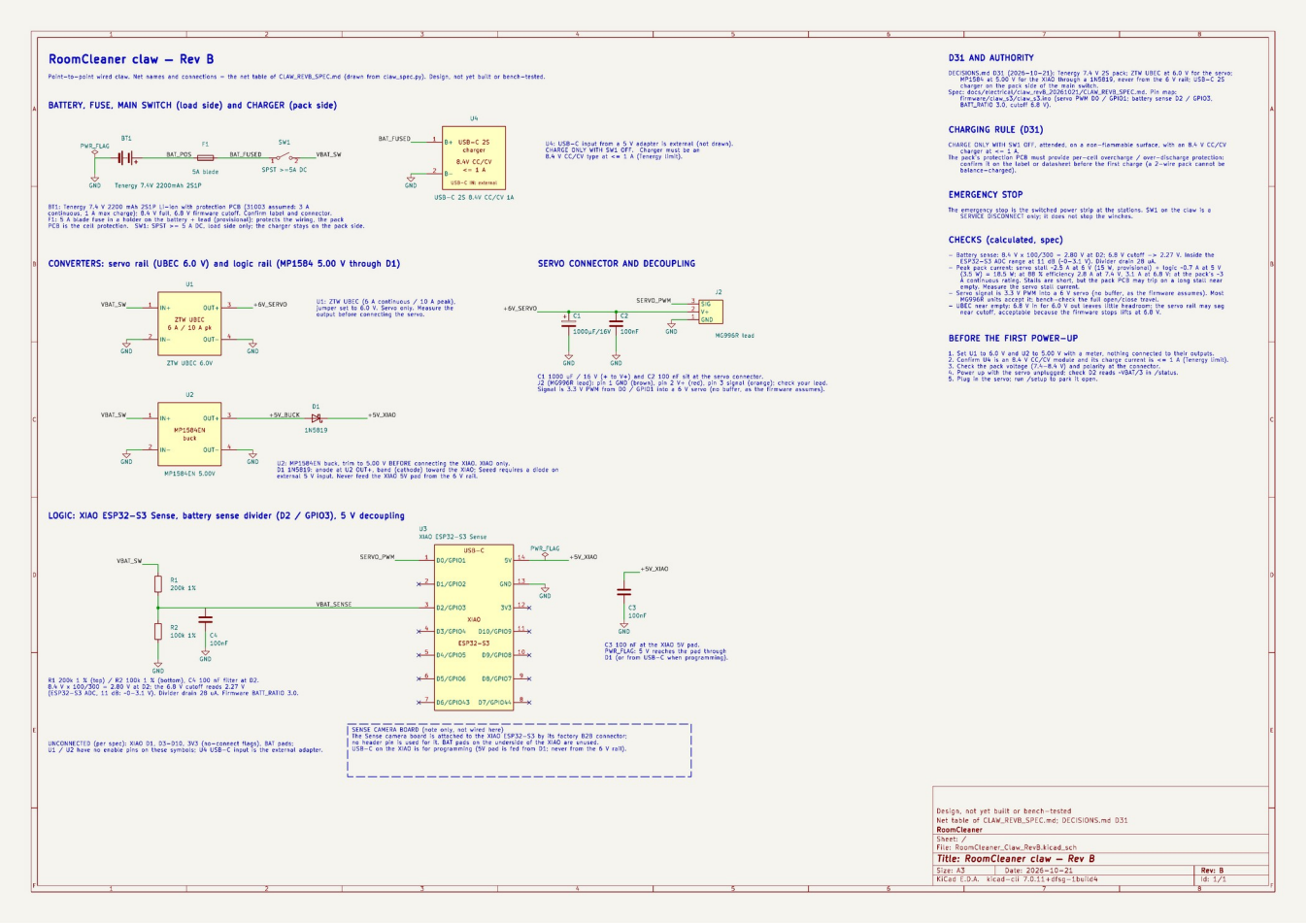
Phone app: home, live claw camera and chat

- Start up: self-check, take up slack, calibrate
- Find: scan from above, detect garments, choose the nearest
- Pick: hover until still, grip, deliver; the camera confirms each step
- Every move waits for approval on the phone; PAUSE holds position

Control Logic: Generation 1 vs. Final

	Generation 1	Final
Where is the claw?	Assumed from the commanded cable lengths	Measured by the claw camera; refit at every startup
Room geometry	Anchors from a tape measure	Anchors and cable lengths fitted, with error bars
Motor control	One central board	A controller per station, coordinated by the laptop
Finding laundry	Ceiling camera	Claw camera
Did the grab work?	Assumed	Camera checks pickup and delivery
Faults	Emergency stop only	PAUSE keeps torque; the emergency stop cuts power and drops the claw

Claw Electrical



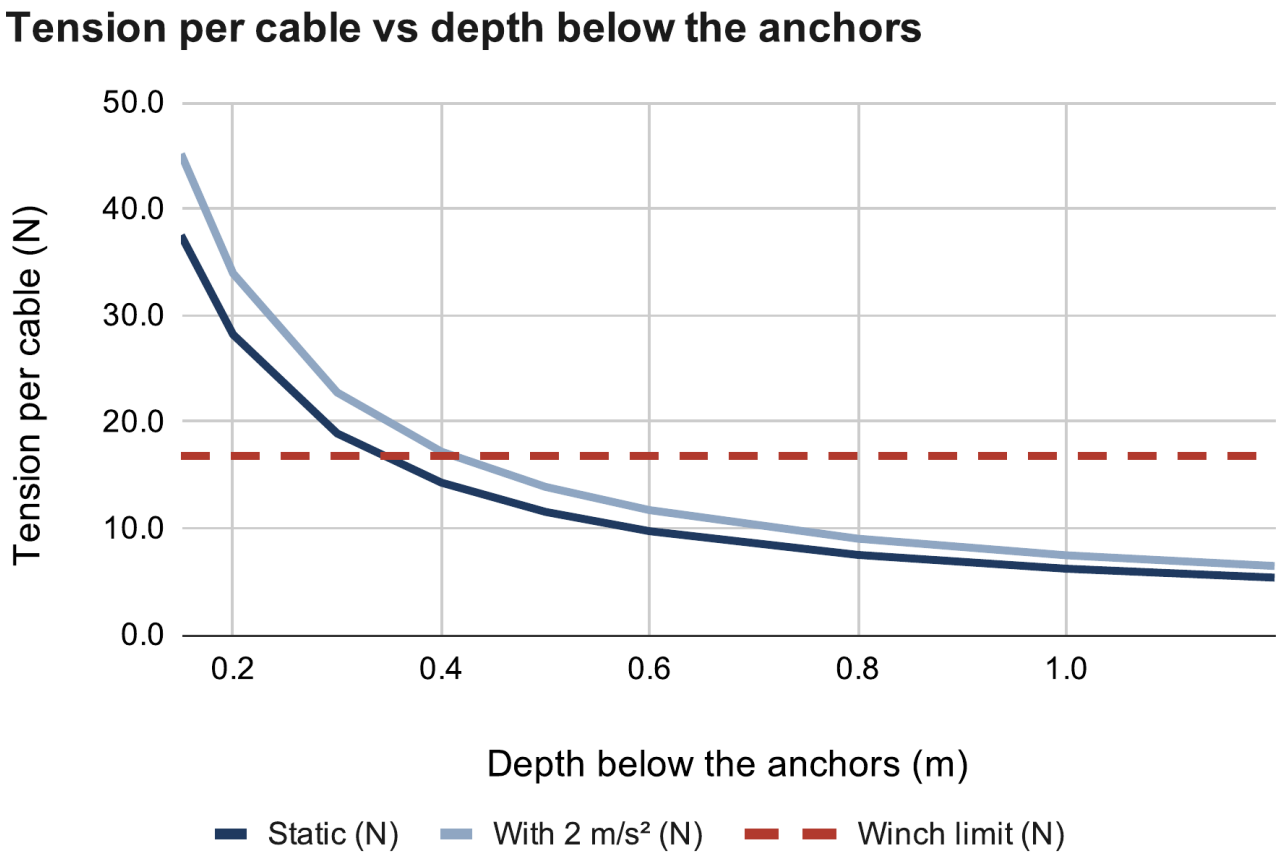
Claw schematic

- Two-cell lithium-ion pack, 7.4 V; 5 A fuse for shorts
- 6 V servo rail and a separate 5 V logic rail
- Peak draw 2.8–3.1 A, at the pack's 3 A rating

Acronyms

Term	Meaning
CAD	Computer-aided design
PETG	Polyethylene terephthalate glycol (the printed plastic)
TPU	Thermoplastic polyurethane (flexible printed plastic)
NEMA 17	National Electrical Manufacturers Association motor frame, 1.7 in square
CNC	Computer numerical control
ESP32	A Wi-Fi microcontroller family (Espressif)
A4988	Stepper-motor driver chip (Allegro)
YOLO	“You Only Look Once”: an object-detection model family
SF	Safety factor: allowable stress divided by actual stress

Calculations and Sources



Calculated (calc sheet: Cable tension)

- All charts come from my engineering calculation sheet
- Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3
- Allegro MicroSystems, A4988 datasheet